XMBA 201A – Managerial Economics

Office Hours Practice Problems: Block 1 Review

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Question 1 — Decision Trees and Reversibility

At the 2009 Frankfurt Motor Show, the founder and CEO of Better Place, Shai Agassi, announced a newly expanded agreement with Renault that commits both companies to a volume of at least 100,000 electric cars in Israel and Denmark by 2016. In May 2011, after his commencement speech at the Berkeley-Haas MBA graduation event, Agassi offered you the job of running the franchise to distribute the needed lithium-ion batteries in Denmark for the next five years (January 2012 through December 2016).

You are now in the market to lease a warehouse outside of Copenhagen.

A small warehouse costs $1 million per year and a large warehouse costs $2 million per year. Current contracts show that your sales will be low in the first two years and that you will only need a small warehouse for these years. Reliable estimates show that there is a 0.3 probability that your sales will take off and that you will have high demand, in which case you will need a large space in years 3 through 5 (and a 0.7 probability that demand stays low and you will only need a small space in all years). If your sales take off and you face high demand, you could increase sales in year 3, but only if you had leased the large warehouse. Data from Better Place includes the following table (which applies to all years):

		Total Annual Revenues	**Annual Variable Costs**
Small Warehouse	Low Demand	3,000,000	1,200,000
	High Demand	3,000,000	1,200,000
Large Warehouse	Low Demand	3,000,000	1,200,000
	High Demand	8,000,000	3,200,000

NOTE: Variable Costs include all costs except the warehouse lease cost. Profit is equal to Total Revenue − Variable Costs − Cost of the Lease. Assume that the interest rate is 0%, so no discounting applies.

a. [10] Draw a decision tree to help you decide whether to lease a large or small warehouse. (Note that you cannot change warehouses, so whatever warehouse you choose you will have in all five years.)

b. [5] Assuming you are a risk-neutral expected-profit maximizing decision maker, should you lease a large warehouse or a small warehouse?

c. [5] Might your answer to part (b) change if you were risk averse?

d. [10] A real estate company offers you an “exit-option lease.” This is a lease for a large warehouse in the first two years, but it would allow, though not require, you to downsize your operation to a small warehouse (less space in the original one) at the end of year 2, after you had learned whether demand would be high or low in years 3 through 5. Each year in the large warehouse would still have an annual cost of $2 million and each year in a small warehouse would still have an annual cost of $1 million. The exit-option lease, however, has two twists. First, it includes a signing fee of $50,000 upfront above and beyond the annual fees. Second, it includes an exercise fee of $500,000, which must be paid at the end of year 2 only if you decide to downsize the warehouse. Assume again that you are risk neutral. Should you sign this exit-option lease instead of the one you chose in part (b)?

Question 2 — Economic Costs and Opportunity Cost

B&B Shampoo buys a special seaweed extract for one of its shampoos from Joe's Seaweed in Rockland, Maine. Joe's makes the seaweed extract specially for B&B, and there are no other known consumers of the product. B&B signed a long-term contract with Joe's and has agreed to buy 1,000 gallons of the extract a month for the next five years at a price of $200 per gallon. B&B overestimated how much seaweed extract it would need, so it currently has a surplus of approximately 300 gallons a month. In part to use the surplus extract, it has decided to launch a styling gel that will use 100 gallons of the extract. What are B&B's economic costs of using the seaweed extract to produce the new styling gel? Explain.

Question 3 — Cost Pass-Through and Profit-Maximizing Pricing

A gas station in a more rural part of the state faces essentially no competition. Its weekly demand is given by $q = 82{,}500 - 30{,}000p$, where p is the price per gallon that it charges. Currently, the gas station is charging $2.63 per gallon. Assume that is the profit-maximizing price. For all intents and purposes, the gas station's marginal cost of selling a gallon of gasoline is the wholesale price it pays per gallon. Suppose, due to recent events, the wholesale price of gasoline rises by 5%.

a. [10] What is the new profit-maximizing price for the gas station to charge its customers?

b. [5] By how much does the gas station's profit per week rise or fall?

c. [5] What does your analysis suggest about the validity of allegations that gas stations get rich when the price of oil rises?